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Fundamentals of Programming

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May 1



Outline

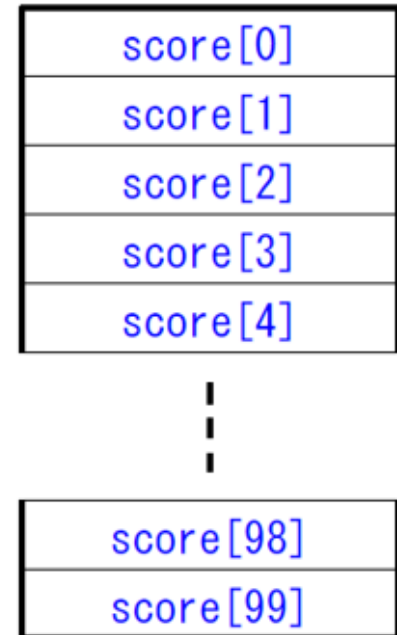
- ◆ Array
- ◆ Char and ASCII Table

Array

- ```
int score[100];
```

int      score[100];  
↓            ↓  
type        number of size
- `score[0]...score[99]`: a block of 100 consecutive elements can be used. (`score[100]` is not available)
- Each element can be used as an int- variable
- The number within `[]` is called subscript
- The subscript can be used as a variable

Size: 100



```
int score[100];

score[10] = 75;
```

=

```
int score[100];
int count;

count = 10;
score[count] = 75;
```

# Exercise 1

```
int st[4]={10, 20, 30, 40};
```

Initialize an array during declaration

```
int st2[]={10, 20, 30};
```

```
Enter 5 integers:
```

```
2
```

```
3
```

```
7
```

```
9
```

```
5
```

```
Displaying integers:
```

```
2
```

```
3
```

```
7
```

```
9
```

```
5
```

Input an array

Output the array

# Examples of Array

```
int st[4]={10, 20, 30, 40};
```

Initialize an array during declaration

```
int st2[]={10, 20, 30};
```

```
#include <stdio.h>

int main() {

 int array[5];

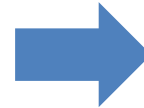
 printf("Enter 5 integers: \n");

 // taking input and storing it in an array
 for(int i = 0; i < 5; i++) {
 scanf("%d", &array[i]);
 }

 printf("Displaying integers: \n");

 // printing elements of an array
 for(int i = 0; i < 5; i++) {
 printf("%d\n", array[i]);
 }

 return 0;
}
```



```
Enter 5 integers:
2
3
7
9
5
Displaying integers:
2
3
7
9
5
```

# Examples of Array (cont.)

```
char a[4]={ 'A' , 'B' , 'C' , '¥0' };
```

In C programming, a string is a sequence of characters terminated with a null character ¥0

## Initialize strings

```
char c[] = "abcd";
```

```
char c[50] = "abcd";
```

```
char c[] = {'a', 'b', 'c', 'd', '\0'};
```

```
char c[5] = {'a', 'b', 'c', 'd', '\0'};
```

| c[0] | c[1] | c[2] | c[3] | c[4] |
|------|------|------|------|------|
| a    | b    | c    | d    | \0   |

# Outline

---

- ◆ Array
- ◆ Char and ASCII Table

# ASCII Table

## ◆ American Standard Code for Information Interchange

### ✦ 0-31: control characters

- Code 13: Ctrl+M, Enter key
- Code 8: Ctrl+H, Backspace

| ASCII Hex Symbol | ASCII Hex Symbol | ASCII Hex Symbol | ASCII Hex Symbol |
|------------------|------------------|------------------|------------------|
| 0 0 NUL          | 16 10 DLE        | 32 20 (space)    | 48 30 0          |
| 1 1 SOH          | 17 11 DC1        | 33 21 !          | 49 31 1          |
| 2 2 STX          | 18 12 DC2        | 34 22 "          | 50 32 2          |
| 3 3 ETX          | 19 13 DC3        | 35 23 #          | 51 33 3          |
| 4 4 EOT          | 20 14 DC4        | 36 24 \$         | 52 34 4          |
| 5 5 ENQ          | 21 15 NAK        | 37 25 %          | 53 35 5          |
| 6 6 ACK          | 22 16 SYN        | 38 26 &          | 54 36 6          |
| 7 7 BEL          | 23 17 ETB        | 39 27 '          | 55 37 7          |
| 8 8 BS           | 24 18 CAN        | 40 28 (          | 56 38 8          |
| 9 9 TAB          | 25 19 EM         | 41 29 )          | 57 39 9          |
| 10 A LF          | 26 1A SUB        | 42 2A *          | 58 3A :          |
| 11 B VT          | 27 1B ESC        | 43 2B +          | 59 3B ;          |
| 12 C FF          | 28 1C FS         | 44 2C ,          | 60 3C <          |
| 13 D CR          | 29 1D GS         | 45 2D -          | 61 3D =          |
| 14 E SO          | 30 1E RS         | 46 2E .          | 62 3E >          |
| 15 F SI          | 31 1F US         | 47 2F /          | 63 3F ?          |



# ASCII Table (cont.)

## ◆ American Standard Code for Information Interchange

❖ 32-126: printable characters

❖ 127: delete code

| ASCII Hex Symbol | ASCII Hex Symbol | ASCII Hex Symbol | ASCII Hex Symbol |
|------------------|------------------|------------------|------------------|
| 64 40 @          | 80 50 P          | 96 60 `          | 112 70 p         |
| 65 41 A          | 81 51 Q          | 97 61 a          | 113 71 q         |
| 66 42 B          | 82 52 R          | 98 62 b          | 114 72 r         |
| 67 43 C          | 83 53 S          | 99 63 c          | 115 73 s         |
| 68 44 D          | 84 54 T          | 100 64 d         | 116 74 t         |
| 69 45 E          | 85 55 U          | 101 65 e         | 117 75 u         |
| 70 46 F          | 86 56 V          | 102 66 f         | 118 76 v         |
| 71 47 G          | 87 57 W          | 103 67 g         | 119 77 w         |
| 72 48 H          | 88 58 X          | 104 68 h         | 120 78 x         |
| 73 49 I          | 89 59 Y          | 105 69 i         | 121 79 y         |
| 74 4A J          | 90 5A Z          | 106 6A j         | 122 7A z         |
| 75 4B K          | 91 5B [          | 107 6B k         | 123 7B {         |
| 76 4C L          | 92 5C \          | 108 6C l         | 124 7C           |
| 77 4D M          | 93 5D ]          | 109 6D m         | 125 7D }         |
| 78 4E N          | 94 5E ^          | 110 6E n         | 126 7E ~         |
| 79 4F O          | 95 5F _          | 111 6F o         | 127 7F           |

# Exercise 2

Try:

```
int main(){
 char a = 71;
 printf("a is %c¥n", a);
}
```

Try:

```
int main(){
 char a = 'F';
 printf("a is %d¥n", a);
}
```

# Exercise 3

## ◆ Use Char for Arithmetic Calculations

“G” + “!” = ?

# Exercise 3

## ◆ Use Char for Arithmetic Calculations

“G” + “!” = ?

```
#include <stdio.h>

int main() {

 char char1 = 'G';
 printf("char1 is %c\n", char1);

 char char2 = '!';
 printf("char2 is %c\n", char2);

 printf("sum is %c\n", char1+char2);

 return 0;

}
```

# Exercise 4

```
int main(){
 char a = 'B';
 a += 32;
 printf("a is %c¥n", a);
}
```

ASCII Hex Symbol

|    |   |     |
|----|---|-----|
| 0  | 0 | NUL |
| 1  | 1 | SOH |
| 2  | 2 | STX |
| 3  | 3 | ETX |
| 4  | 4 | EOT |
| 5  | 5 | ENQ |
| 6  | 6 | ACK |
| 7  | 7 | BEL |
| 8  | 8 | BS  |
| 9  | 9 | TAB |
| 10 | A | LF  |
| 11 | B | VT  |
| 12 | C | FF  |
| 13 | D | CR  |
| 14 | E | SO  |
| 15 | F | SI  |

ASCII Hex Symbol

|    |    |     |
|----|----|-----|
| 16 | 10 | DLE |
| 17 | 11 | DC1 |
| 18 | 12 | DC2 |
| 19 | 13 | DC3 |
| 20 | 14 | DC4 |
| 21 | 15 | NAK |
| 22 | 16 | SYN |
| 23 | 17 | ETB |
| 24 | 18 | CAN |
| 25 | 19 | EM  |
| 26 | 1A | SUB |
| 27 | 1B | ESC |
| 28 | 1C | FS  |
| 29 | 1D | GS  |
| 30 | 1E | RS  |
| 31 | 1F | US  |

ASCII Hex Symbol

|    |    |         |
|----|----|---------|
| 32 | 20 | (space) |
| 33 | 21 | !       |
| 34 | 22 | "       |
| 35 | 23 | #       |
| 36 | 24 | \$      |
| 37 | 25 | %       |
| 38 | 26 | &       |
| 39 | 27 | '       |
| 40 | 28 | (       |
| 41 | 29 | )       |
| 42 | 2A | *       |
| 43 | 2B | +       |
| 44 | 2C | ,       |
| 45 | 2D | -       |
| 46 | 2E | .       |
| 47 | 2F | /       |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 48 | 30 | 0 |
| 49 | 31 | 1 |
| 50 | 32 | 2 |
| 51 | 33 | 3 |
| 52 | 34 | 4 |
| 53 | 35 | 5 |
| 54 | 36 | 6 |
| 55 | 37 | 7 |
| 56 | 38 | 8 |
| 57 | 39 | 9 |
| 58 | 3A | : |
| 59 | 3B | ; |
| 60 | 3C | < |
| 61 | 3D | = |
| 62 | 3E | > |
| 63 | 3F | ? |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 64 | 40 | @ |
| 65 | 41 | A |
| 66 | 42 | B |
| 67 | 43 | C |
| 68 | 44 | D |
| 69 | 45 | E |
| 70 | 46 | F |
| 71 | 47 | G |
| 72 | 48 | H |
| 73 | 49 | I |
| 74 | 4A | J |
| 75 | 4B | K |
| 76 | 4C | L |
| 77 | 4D | M |
| 78 | 4E | N |
| 79 | 4F | O |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 80 | 50 | P |
| 81 | 51 | Q |
| 82 | 52 | R |
| 83 | 53 | S |
| 84 | 54 | T |
| 85 | 55 | U |
| 86 | 56 | V |
| 87 | 57 | W |
| 88 | 58 | X |
| 89 | 59 | Y |
| 90 | 5A | Z |
| 91 | 5B | [ |
| 92 | 5C | \ |
| 93 | 5D | ] |
| 94 | 5E | ^ |
| 95 | 5F | _ |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 96  | 60 | ` |
| 97  | 61 | a |
| 98  | 62 | b |
| 99  | 63 | c |
| 100 | 64 | d |
| 101 | 65 | e |
| 102 | 66 | f |
| 103 | 67 | g |
| 104 | 68 | h |
| 105 | 69 | i |
| 106 | 6A | j |
| 107 | 6B | k |
| 108 | 6C | l |
| 109 | 6D | m |
| 110 | 6E | n |
| 111 | 6F | o |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 112 | 70 | p |
| 113 | 71 | q |
| 114 | 72 | r |
| 115 | 73 | s |
| 116 | 74 | t |
| 117 | 75 | u |
| 118 | 76 | v |
| 119 | 77 | w |
| 120 | 78 | x |
| 121 | 79 | y |
| 122 | 7A | z |
| 123 | 7B | { |
| 124 | 7C |   |
| 125 | 7D | } |
| 126 | 7E | ~ |
| 127 | 7F |   |

# Efficiency Conversion between Char and Int

Int -> char

ASCII Hex Symbol

|    |   |     |
|----|---|-----|
| 0  | 0 | NUL |
| 1  | 1 | SOH |
| 2  | 2 | STX |
| 3  | 3 | ETX |
| 4  | 4 | EOT |
| 5  | 5 | ENQ |
| 6  | 6 | ACK |
| 7  | 7 | BEL |
| 8  | 8 | BS  |
| 9  | 9 | TAB |
| 10 | A | LF  |
| 11 | B | VT  |
| 12 | C | FF  |
| 13 | D | CR  |
| 14 | E | SO  |
| 15 | F | SI  |

ASCII Hex Symbol

|    |    |     |
|----|----|-----|
| 16 | 10 | DLE |
| 17 | 11 | DC1 |
| 18 | 12 | DC2 |
| 19 | 13 | DC3 |
| 20 | 14 | DC4 |
| 21 | 15 | NAK |
| 22 | 16 | SYN |
| 23 | 17 | ETB |
| 24 | 18 | CAN |
| 25 | 19 | EM  |
| 26 | 1A | SUB |
| 27 | 1B | ESC |
| 28 | 1C | FS  |
| 29 | 1D | GS  |
| 30 | 1E | RS  |
| 31 | 1F | US  |

ASCII Hex Symbol

|    |    |         |
|----|----|---------|
| 32 | 20 | (space) |
| 33 | 21 | !       |
| 34 | 22 | "       |
| 35 | 23 | #       |
| 36 | 24 | \$      |
| 37 | 25 | %       |
| 38 | 26 | &       |
| 39 | 27 | '       |
| 40 | 28 | (       |
| 41 | 29 | )       |
| 42 | 2A | *       |
| 43 | 2B | +       |
| 44 | 2C | ,       |
| 45 | 2D | -       |
| 46 | 2E | .       |
| 47 | 2F | /       |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 48 | 30 | 0 |
| 49 | 31 | 1 |
| 50 | 32 | 2 |
| 51 | 33 | 3 |
| 52 | 34 | 4 |
| 53 | 35 | 5 |
| 54 | 36 | 6 |
| 55 | 37 | 7 |
| 56 | 38 | 8 |
| 57 | 39 | 9 |
| 58 | 3A | : |
| 59 | 3B | ; |
| 60 | 3C | < |
| 61 | 3D | = |
| 62 | 3E | > |
| 63 | 3F | ? |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 64 | 40 | @ |
| 65 | 41 | A |
| 66 | 42 | B |
| 67 | 43 | C |
| 68 | 44 | D |
| 69 | 45 | E |
| 70 | 46 | F |
| 71 | 47 | G |
| 72 | 48 | H |
| 73 | 49 | I |
| 74 | 4A | J |
| 75 | 4B | K |
| 76 | 4C | L |
| 77 | 4D | M |
| 78 | 4E | N |
| 79 | 4F | O |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 80 | 50 | P |
| 81 | 51 | Q |
| 82 | 52 | R |
| 83 | 53 | S |
| 84 | 54 | T |
| 85 | 55 | U |
| 86 | 56 | V |
| 87 | 57 | W |
| 88 | 58 | X |
| 89 | 59 | Y |
| 90 | 5A | Z |
| 91 | 5B | [ |
| 92 | 5C | \ |
| 93 | 5D | ] |
| 94 | 5E | ^ |
| 95 | 5F | _ |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 96  | 60 | ` |
| 97  | 61 | a |
| 98  | 62 | b |
| 99  | 63 | c |
| 100 | 64 | d |
| 101 | 65 | e |
| 102 | 66 | f |
| 103 | 67 | g |
| 104 | 68 | h |
| 105 | 69 | i |
| 106 | 6A | j |
| 107 | 6B | k |
| 108 | 6C | l |
| 109 | 6D | m |
| 110 | 6E | n |
| 111 | 6F | o |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 112 | 70 | p |
| 113 | 71 | q |
| 114 | 72 | r |
| 115 | 73 | s |
| 116 | 74 | t |
| 117 | 75 | u |
| 118 | 76 | v |
| 119 | 77 | w |
| 120 | 78 | x |
| 121 | 79 | y |
| 122 | 7A | z |
| 123 | 7B | { |
| 124 | 7C |   |
| 125 | 7D | } |
| 126 | 7E | ~ |
| 127 | 7F |   |

# Efficiency Conversion between Char and Int

Int -> char

```
#include <stdio.h>

int main() {

 int i = 5;

 char a = i+'0';

 printf("a is %d\n", a);
 printf("a is %c\n", a);

 return 0;
}
```

ASCII Hex Symbol

|    |   |     |
|----|---|-----|
| 0  | 0 | NUL |
| 1  | 1 | SOH |
| 2  | 2 | STX |
| 3  | 3 | ETX |
| 4  | 4 | EOT |
| 5  | 5 | ENQ |
| 6  | 6 | ACK |
| 7  | 7 | BEL |
| 8  | 8 | BS  |
| 9  | 9 | TAB |
| 10 | A | LF  |
| 11 | B | VT  |
| 12 | C | FF  |
| 13 | D | CR  |
| 14 | E | SO  |
| 15 | F | SI  |

ASCII Hex Symbol

|    |    |     |
|----|----|-----|
| 16 | 10 | DLE |
| 17 | 11 | DC1 |
| 18 | 12 | DC2 |
| 19 | 13 | DC3 |
| 20 | 14 | DC4 |
| 21 | 15 | NAK |
| 22 | 16 | SYN |
| 23 | 17 | ETB |
| 24 | 18 | CAN |
| 25 | 19 | EM  |
| 26 | 1A | SUB |
| 27 | 1B | ESC |
| 28 | 1C | FS  |
| 29 | 1D | GS  |
| 30 | 1E | RS  |
| 31 | 1F | US  |

ASCII Hex Symbol

|    |    |         |
|----|----|---------|
| 32 | 20 | (space) |
| 33 | 21 | !       |
| 34 | 22 | "       |
| 35 | 23 | #       |
| 36 | 24 | \$      |
| 37 | 25 | %       |
| 38 | 26 | &       |
| 39 | 27 | '       |
| 40 | 28 | (       |
| 41 | 29 | )       |
| 42 | 2A | *       |
| 43 | 2B | +       |
| 44 | 2C | ,       |
| 45 | 2D | -       |
| 46 | 2E | .       |
| 47 | 2F | /       |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 48 | 30 | 0 |
| 49 | 31 | 1 |
| 50 | 32 | 2 |
| 51 | 33 | 3 |
| 52 | 34 | 4 |
| 53 | 35 | 5 |
| 54 | 36 | 6 |
| 55 | 37 | 7 |
| 56 | 38 | 8 |
| 57 | 39 | 9 |
| 58 | 3A | : |
| 59 | 3B | ; |
| 60 | 3C | < |
| 61 | 3D | = |
| 62 | 3E | > |
| 63 | 3F | ? |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 64 | 40 | @ |
| 65 | 41 | A |
| 66 | 42 | B |
| 67 | 43 | C |
| 68 | 44 | D |
| 69 | 45 | E |
| 70 | 46 | F |
| 71 | 47 | G |
| 72 | 48 | H |
| 73 | 49 | I |
| 74 | 4A | J |
| 75 | 4B | K |
| 76 | 4C | L |
| 77 | 4D | M |
| 78 | 4E | N |
| 79 | 4F | O |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 80 | 50 | P |
| 81 | 51 | Q |
| 82 | 52 | R |
| 83 | 53 | S |
| 84 | 54 | T |
| 85 | 55 | U |
| 86 | 56 | V |
| 87 | 57 | W |
| 88 | 58 | X |
| 89 | 59 | Y |
| 90 | 5A | Z |
| 91 | 5B | [ |
| 92 | 5C | \ |
| 93 | 5D | ] |
| 94 | 5E | ^ |
| 95 | 5F | _ |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 96  | 60 | ` |
| 97  | 61 | a |
| 98  | 62 | b |
| 99  | 63 | c |
| 100 | 64 | d |
| 101 | 65 | e |
| 102 | 66 | f |
| 103 | 67 | g |
| 104 | 68 | h |
| 105 | 69 | i |
| 106 | 6A | j |
| 107 | 6B | k |
| 108 | 6C | l |
| 109 | 6D | m |
| 110 | 6E | n |
| 111 | 6F | o |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 112 | 70 | p |
| 113 | 71 | q |
| 114 | 72 | r |
| 115 | 73 | s |
| 116 | 74 | t |
| 117 | 75 | u |
| 118 | 76 | v |
| 119 | 77 | w |
| 120 | 78 | x |
| 121 | 79 | y |
| 122 | 7A | z |
| 123 | 7B | { |
| 124 | 7C |   |
| 125 | 7D | } |
| 126 | 7E | ~ |
| 127 | 7F |   |

# Efficiency Conversion between Char and Int

Char -> int

ASCII Hex Symbol

|    |   |     |
|----|---|-----|
| 0  | 0 | NUL |
| 1  | 1 | SOH |
| 2  | 2 | STX |
| 3  | 3 | ETX |
| 4  | 4 | EOT |
| 5  | 5 | ENQ |
| 6  | 6 | ACK |
| 7  | 7 | BEL |
| 8  | 8 | BS  |
| 9  | 9 | TAB |
| 10 | A | LF  |
| 11 | B | VT  |
| 12 | C | FF  |
| 13 | D | CR  |
| 14 | E | SO  |
| 15 | F | SI  |

ASCII Hex Symbol

|    |    |     |
|----|----|-----|
| 16 | 10 | DLE |
| 17 | 11 | DC1 |
| 18 | 12 | DC2 |
| 19 | 13 | DC3 |
| 20 | 14 | DC4 |
| 21 | 15 | NAK |
| 22 | 16 | SYN |
| 23 | 17 | ETB |
| 24 | 18 | CAN |
| 25 | 19 | EM  |
| 26 | 1A | SUB |
| 27 | 1B | ESC |
| 28 | 1C | FS  |
| 29 | 1D | GS  |
| 30 | 1E | RS  |
| 31 | 1F | US  |

ASCII Hex Symbol

|    |    |         |
|----|----|---------|
| 32 | 20 | (space) |
| 33 | 21 | !       |
| 34 | 22 | "       |
| 35 | 23 | #       |
| 36 | 24 | \$      |
| 37 | 25 | %       |
| 38 | 26 | &       |
| 39 | 27 | '       |
| 40 | 28 | (       |
| 41 | 29 | )       |
| 42 | 2A | *       |
| 43 | 2B | +       |
| 44 | 2C | ,       |
| 45 | 2D | -       |
| 46 | 2E | .       |
| 47 | 2F | /       |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 48 | 30 | 0 |
| 49 | 31 | 1 |
| 50 | 32 | 2 |
| 51 | 33 | 3 |
| 52 | 34 | 4 |
| 53 | 35 | 5 |
| 54 | 36 | 6 |
| 55 | 37 | 7 |
| 56 | 38 | 8 |
| 57 | 39 | 9 |
| 58 | 3A | : |
| 59 | 3B | ; |
| 60 | 3C | < |
| 61 | 3D | = |
| 62 | 3E | > |
| 63 | 3F | ? |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 64 | 40 | @ |
| 65 | 41 | A |
| 66 | 42 | B |
| 67 | 43 | C |
| 68 | 44 | D |
| 69 | 45 | E |
| 70 | 46 | F |
| 71 | 47 | G |
| 72 | 48 | H |
| 73 | 49 | I |
| 74 | 4A | J |
| 75 | 4B | K |
| 76 | 4C | L |
| 77 | 4D | M |
| 78 | 4E | N |
| 79 | 4F | O |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 80 | 50 | P |
| 81 | 51 | Q |
| 82 | 52 | R |
| 83 | 53 | S |
| 84 | 54 | T |
| 85 | 55 | U |
| 86 | 56 | V |
| 87 | 57 | W |
| 88 | 58 | X |
| 89 | 59 | Y |
| 90 | 5A | Z |
| 91 | 5B | [ |
| 92 | 5C | \ |
| 93 | 5D | ] |
| 94 | 5E | ^ |
| 95 | 5F | _ |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 96  | 60 | ` |
| 97  | 61 | a |
| 98  | 62 | b |
| 99  | 63 | c |
| 100 | 64 | d |
| 101 | 65 | e |
| 102 | 66 | f |
| 103 | 67 | g |
| 104 | 68 | h |
| 105 | 69 | i |
| 106 | 6A | j |
| 107 | 6B | k |
| 108 | 6C | l |
| 109 | 6D | m |
| 110 | 6E | n |
| 111 | 6F | o |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 112 | 70 | p |
| 113 | 71 | q |
| 114 | 72 | r |
| 115 | 73 | s |
| 116 | 74 | t |
| 117 | 75 | u |
| 118 | 76 | v |
| 119 | 77 | w |
| 120 | 78 | x |
| 121 | 79 | y |
| 122 | 7A | z |
| 123 | 7B | { |
| 124 | 7C |   |
| 125 | 7D | } |
| 126 | 7E | ~ |
| 127 | 7F |   |



# Efficiency Conversion between Char and Int

Char -> int

```
#include <stdio.h>

int main() {

 char i = '5';

 int a = i-'0';

 printf("a is %d\n", a);

 return 0;
}
```

ASCII Hex Symbol

|    |   |     |
|----|---|-----|
| 0  | 0 | NUL |
| 1  | 1 | SOH |
| 2  | 2 | STX |
| 3  | 3 | ETX |
| 4  | 4 | EOT |
| 5  | 5 | ENQ |
| 6  | 6 | ACK |
| 7  | 7 | BEL |
| 8  | 8 | BS  |
| 9  | 9 | TAB |
| 10 | A | LF  |
| 11 | B | VT  |
| 12 | C | FF  |
| 13 | D | CR  |
| 14 | E | SO  |
| 15 | F | SI  |

ASCII Hex Symbol

|    |    |     |
|----|----|-----|
| 16 | 10 | DLE |
| 17 | 11 | DC1 |
| 18 | 12 | DC2 |
| 19 | 13 | DC3 |
| 20 | 14 | DC4 |
| 21 | 15 | NAK |
| 22 | 16 | SYN |
| 23 | 17 | ETB |
| 24 | 18 | CAN |
| 25 | 19 | EM  |
| 26 | 1A | SUB |
| 27 | 1B | ESC |
| 28 | 1C | FS  |
| 29 | 1D | GS  |
| 30 | 1E | RS  |
| 31 | 1F | US  |

ASCII Hex Symbol

|    |    |         |
|----|----|---------|
| 32 | 20 | (space) |
| 33 | 21 | !       |
| 34 | 22 | "       |
| 35 | 23 | #       |
| 36 | 24 | \$      |
| 37 | 25 | %       |
| 38 | 26 | &       |
| 39 | 27 | '       |
| 40 | 28 | (       |
| 41 | 29 | )       |
| 42 | 2A | *       |
| 43 | 2B | +       |
| 44 | 2C | ,       |
| 45 | 2D | -       |
| 46 | 2E | .       |
| 47 | 2F | /       |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 48 | 30 | 0 |
| 49 | 31 | 1 |
| 50 | 32 | 2 |
| 51 | 33 | 3 |
| 52 | 34 | 4 |
| 53 | 35 | 5 |
| 54 | 36 | 6 |
| 55 | 37 | 7 |
| 56 | 38 | 8 |
| 57 | 39 | 9 |
| 58 | 3A | : |
| 59 | 3B | ; |
| 60 | 3C | < |
| 61 | 3D | = |
| 62 | 3E | > |
| 63 | 3F | ? |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 64 | 40 | @ |
| 65 | 41 | A |
| 66 | 42 | B |
| 67 | 43 | C |
| 68 | 44 | D |
| 69 | 45 | E |
| 70 | 46 | F |
| 71 | 47 | G |
| 72 | 48 | H |
| 73 | 49 | I |
| 74 | 4A | J |
| 75 | 4B | K |
| 76 | 4C | L |
| 77 | 4D | M |
| 78 | 4E | N |
| 79 | 4F | O |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 80 | 50 | P |
| 81 | 51 | Q |
| 82 | 52 | R |
| 83 | 53 | S |
| 84 | 54 | T |
| 85 | 55 | U |
| 86 | 56 | V |
| 87 | 57 | W |
| 88 | 58 | X |
| 89 | 59 | Y |
| 90 | 5A | Z |
| 91 | 5B | [ |
| 92 | 5C | \ |
| 93 | 5D | ] |
| 94 | 5E | ^ |
| 95 | 5F | _ |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 96  | 60 | ` |
| 97  | 61 | a |
| 98  | 62 | b |
| 99  | 63 | c |
| 100 | 64 | d |
| 101 | 65 | e |
| 102 | 66 | f |
| 103 | 67 | g |
| 104 | 68 | h |
| 105 | 69 | i |
| 106 | 6A | j |
| 107 | 6B | k |
| 108 | 6C | l |
| 109 | 6D | m |
| 110 | 6E | n |
| 111 | 6F | o |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 112 | 70 | p |
| 113 | 71 | q |
| 114 | 72 | r |
| 115 | 73 | s |
| 116 | 74 | t |
| 117 | 75 | u |
| 118 | 76 | v |
| 119 | 77 | w |
| 120 | 78 | x |
| 121 | 79 | y |
| 122 | 7A | z |
| 123 | 7B | { |
| 124 | 7C |   |
| 125 | 7D | } |
| 126 | 7E | ~ |
| 127 | 7F |   |

# Exercise 5

- ◆ Write a program to print alphabets a-z as follow



ASCII Hex Symbol

|    |   |     |
|----|---|-----|
| 0  | 0 | NUL |
| 1  | 1 | SOH |
| 2  | 2 | STX |
| 3  | 3 | ETX |
| 4  | 4 | EOT |
| 5  | 5 | ENQ |
| 6  | 6 | ACK |
| 7  | 7 | BEL |
| 8  | 8 | BS  |
| 9  | 9 | TAB |
| 10 | A | LF  |
| 11 | B | VT  |
| 12 | C | FF  |
| 13 | D | CR  |
| 14 | E | SO  |
| 15 | F | SI  |

ASCII Hex Symbol

|    |    |     |
|----|----|-----|
| 16 | 10 | DLE |
| 17 | 11 | DC1 |
| 18 | 12 | DC2 |
| 19 | 13 | DC3 |
| 20 | 14 | DC4 |
| 21 | 15 | NAK |
| 22 | 16 | SYN |
| 23 | 17 | ETB |
| 24 | 18 | CAN |
| 25 | 19 | EM  |
| 26 | 1A | SUB |
| 27 | 1B | ESC |
| 28 | 1C | FS  |
| 29 | 1D | GS  |
| 30 | 1E | RS  |
| 31 | 1F | US  |

ASCII Hex Symbol

|    |    |         |
|----|----|---------|
| 32 | 20 | (space) |
| 33 | 21 | !       |
| 34 | 22 | "       |
| 35 | 23 | #       |
| 36 | 24 | \$      |
| 37 | 25 | %       |
| 38 | 26 | &       |
| 39 | 27 | '       |
| 40 | 28 | (       |
| 41 | 29 | )       |
| 42 | 2A | *       |
| 43 | 2B | +       |
| 44 | 2C | ,       |
| 45 | 2D | -       |
| 46 | 2E | .       |
| 47 | 2F | /       |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 48 | 30 | 0 |
| 49 | 31 | 1 |
| 50 | 32 | 2 |
| 51 | 33 | 3 |
| 52 | 34 | 4 |
| 53 | 35 | 5 |
| 54 | 36 | 6 |
| 55 | 37 | 7 |
| 56 | 38 | 8 |
| 57 | 39 | 9 |
| 58 | 3A | : |
| 59 | 3B | ; |
| 60 | 3C | < |
| 61 | 3D | = |
| 62 | 3E | > |
| 63 | 3F | ? |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 64 | 40 | @ |
| 65 | 41 | A |
| 66 | 42 | B |
| 67 | 43 | C |
| 68 | 44 | D |
| 69 | 45 | E |
| 70 | 46 | F |
| 71 | 47 | G |
| 72 | 48 | H |
| 73 | 49 | I |
| 74 | 4A | J |
| 75 | 4B | K |
| 76 | 4C | L |
| 77 | 4D | M |
| 78 | 4E | N |
| 79 | 4F | O |

ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 80 | 50 | P |
| 81 | 51 | Q |
| 82 | 52 | R |
| 83 | 53 | S |
| 84 | 54 | T |
| 85 | 55 | U |
| 86 | 56 | V |
| 87 | 57 | W |
| 88 | 58 | X |
| 89 | 59 | Y |
| 90 | 5A | Z |
| 91 | 5B | [ |
| 92 | 5C | \ |
| 93 | 5D | ] |
| 94 | 5E | ^ |
| 95 | 5F | _ |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 96  | 60 | ` |
| 97  | 61 | a |
| 98  | 62 | b |
| 99  | 63 | c |
| 100 | 64 | d |
| 101 | 65 | e |
| 102 | 66 | f |
| 103 | 67 | g |
| 104 | 68 | h |
| 105 | 69 | i |
| 106 | 6A | j |
| 107 | 6B | k |
| 108 | 6C | l |
| 109 | 6D | m |
| 110 | 6E | n |
| 111 | 6F | o |

ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 112 | 70 | p |
| 113 | 71 | q |
| 114 | 72 | r |
| 115 | 73 | s |
| 116 | 74 | t |
| 117 | 75 | u |
| 118 | 76 | v |
| 119 | 77 | w |
| 120 | 78 | x |
| 121 | 79 | y |
| 122 | 7A | z |
| 123 | 7B | { |
| 124 | 7C |   |
| 125 | 7D | } |
| 126 | 7E | ~ |
| 127 | 7F |   |

# Exercise 5 (cont.)

```
#include <stdio.h>

int main() {

 for(char i=97; i<=122; i++)
 {
 printf("%c\n",i);
 }

 return 0;
}
```

## ASCII Hex Symbol

|    |   |     |
|----|---|-----|
| 0  | 0 | NUL |
| 1  | 1 | SOH |
| 2  | 2 | STX |
| 3  | 3 | ETX |
| 4  | 4 | EOT |
| 5  | 5 | ENQ |
| 6  | 6 | ACK |
| 7  | 7 | BEL |
| 8  | 8 | BS  |
| 9  | 9 | TAB |
| 10 | A | LF  |
| 11 | B | VT  |
| 12 | C | FF  |
| 13 | D | CR  |
| 14 | E | SO  |
| 15 | F | SI  |

## ASCII Hex Symbol

|    |    |     |
|----|----|-----|
| 16 | 10 | DLE |
| 17 | 11 | DC1 |
| 18 | 12 | DC2 |
| 19 | 13 | DC3 |
| 20 | 14 | DC4 |
| 21 | 15 | NAK |
| 22 | 16 | SYN |
| 23 | 17 | ETB |
| 24 | 18 | CAN |
| 25 | 19 | EM  |
| 26 | 1A | SUB |
| 27 | 1B | ESC |
| 28 | 1C | FS  |
| 29 | 1D | GS  |
| 30 | 1E | RS  |
| 31 | 1F | US  |

## ASCII Hex Symbol

|    |    |         |
|----|----|---------|
| 32 | 20 | (space) |
| 33 | 21 | !       |
| 34 | 22 | "       |
| 35 | 23 | #       |
| 36 | 24 | \$      |
| 37 | 25 | %       |
| 38 | 26 | &       |
| 39 | 27 | '       |
| 40 | 28 | (       |
| 41 | 29 | )       |
| 42 | 2A | *       |
| 43 | 2B | +       |
| 44 | 2C | ,       |
| 45 | 2D | -       |
| 46 | 2E | .       |
| 47 | 2F | /       |

## ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 48 | 30 | 0 |
| 49 | 31 | 1 |
| 50 | 32 | 2 |
| 51 | 33 | 3 |
| 52 | 34 | 4 |
| 53 | 35 | 5 |
| 54 | 36 | 6 |
| 55 | 37 | 7 |
| 56 | 38 | 8 |
| 57 | 39 | 9 |
| 58 | 3A | : |
| 59 | 3B | ; |
| 60 | 3C | < |
| 61 | 3D | = |
| 62 | 3E | > |
| 63 | 3F | ? |

## ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 64 | 40 | @ |
| 65 | 41 | A |
| 66 | 42 | B |
| 67 | 43 | C |
| 68 | 44 | D |
| 69 | 45 | E |
| 70 | 46 | F |
| 71 | 47 | G |
| 72 | 48 | H |
| 73 | 49 | I |
| 74 | 4A | J |
| 75 | 4B | K |
| 76 | 4C | L |
| 77 | 4D | M |
| 78 | 4E | N |
| 79 | 4F | O |

## ASCII Hex Symbol

|    |    |   |
|----|----|---|
| 80 | 50 | P |
| 81 | 51 | Q |
| 82 | 52 | R |
| 83 | 53 | S |
| 84 | 54 | T |
| 85 | 55 | U |
| 86 | 56 | V |
| 87 | 57 | W |
| 88 | 58 | X |
| 89 | 59 | Y |
| 90 | 5A | Z |
| 91 | 5B | [ |
| 92 | 5C | \ |
| 93 | 5D | ] |
| 94 | 5E | ^ |
| 95 | 5F | _ |

## ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 96  | 60 | ` |
| 97  | 61 | a |
| 98  | 62 | b |
| 99  | 63 | c |
| 100 | 64 | d |
| 101 | 65 | e |
| 102 | 66 | f |
| 103 | 67 | g |
| 104 | 68 | h |
| 105 | 69 | i |
| 106 | 6A | j |
| 107 | 6B | k |
| 108 | 6C | l |
| 109 | 6D | m |
| 110 | 6E | n |
| 111 | 6F | o |

## ASCII Hex Symbol

|     |    |   |
|-----|----|---|
| 112 | 70 | p |
| 113 | 71 | q |
| 114 | 72 | r |
| 115 | 73 | s |
| 116 | 74 | t |
| 117 | 75 | u |
| 118 | 76 | v |
| 119 | 77 | w |
| 120 | 78 | x |
| 121 | 79 | y |
| 122 | 7A | z |
| 123 | 7B | { |
| 124 | 7C |   |
| 125 | 7D | } |
| 126 | 7E | ~ |
| 127 | 7F |   |

# Exercise 6

- ◆ Please change the small letters to big letters (keep the big letters)
- ❖ For example, change {'i','n','T','E','r'} to {'I','N','T','E','R'}

| ASCII Hex Symbol |    |   | ASCII Hex Symbol |    |   | ASCII Hex Symbol |    |   | ASCII Hex Symbol |    |   |
|------------------|----|---|------------------|----|---|------------------|----|---|------------------|----|---|
| 64               | 40 | @ | 80               | 50 | P | 96               | 60 | ` | 112              | 70 | p |
| 65               | 41 | A | 81               | 51 | Q | 97               | 61 | a | 113              | 71 | q |
| 66               | 42 | B | 82               | 52 | R | 98               | 62 | b | 114              | 72 | r |
| 67               | 43 | C | 83               | 53 | S | 99               | 63 | c | 115              | 73 | s |
| 68               | 44 | D | 84               | 54 | T | 100              | 64 | d | 116              | 74 | t |
| 69               | 45 | E | 85               | 55 | U | 101              | 65 | e | 117              | 75 | u |
| 70               | 46 | F | 86               | 56 | V | 102              | 66 | f | 118              | 76 | v |
| 71               | 47 | G | 87               | 57 | W | 103              | 67 | g | 119              | 77 | w |
| 72               | 48 | H | 88               | 58 | X | 104              | 68 | h | 120              | 78 | x |
| 73               | 49 | I | 89               | 59 | Y | 105              | 69 | i | 121              | 79 | y |
| 74               | 4A | J | 90               | 5A | Z | 106              | 6A | j | 122              | 7A | z |
| 75               | 4B | K | 91               | 5B | [ | 107              | 6B | k | 123              | 7B | { |
| 76               | 4C | L | 92               | 5C | \ | 108              | 6C | l | 124              | 7C |   |
| 77               | 4D | M | 93               | 5D | ] | 109              | 6D | m | 125              | 7D | } |
| 78               | 4E | N | 94               | 5E | ^ | 110              | 6E | n | 126              | 7E | ~ |
| 79               | 4F | O | 95               | 5F | _ | 111              | 6F | o | 127              | 7F |   |